

Clinical Decision Making and Pattern Recognition in Health Care:

Agentic Generative AI for Treatment, Payment, and Operations (TPO)

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1. Concept Definition

Clinical decision-making is the computational and cognitive process of evaluating patient data to select diagnoses, treatments, and interventions. Pattern recognition, the identification of clinically significant signals within high-dimensional datasets, is the foundation of this process. Historically a physician-led task, it is now increasingly augmented by Agentic Generative AI systems capable of multi-step autonomous reasoning.

Unlike standard LLMs that respond to isolated queries, agentic systems are goal-directed: they retrieve evidence, reconcile conflicting data, and execute auditable decisions. In the context of Cotiviti's mandate, an agentic system does not merely predict risk; it cross-references ML predictions against clinical guidelines and produces grounded, citable recommendations across Treatment (T), Payment (P), and Operations (O). The proof-of-concept accompanying this report demonstrates this architecture using a diabetes risk prediction ensemble paired with a SHAP explainability layer and a Llama-3.3-70B reasoning agent grounded in 2023 ADA guidelines.

2. Relevant Trends

2.1 Market Acceleration

The Healthcare Agentic AI market reached \$0.7 billion in 2025 and is forecast to grow at a 44.83% CAGR to \$4.46 billion by 2030. Clinical Decision Support captured 35.2% of 2024 revenue, while payer and insurance applications are projected to grow at the highest CAGR, 35.2%, a segment directly aligned with Cotiviti's business model. LLM-based agents are forecast to grow at a 47.2% CAGR, signaling a structural shift from static models to reasoning-capable systems (Mordor Intelligence, 2026).

2.2 From Generative to Agentic

The sector has moved beyond experimental chatbot deployments to autonomous, goal-directed systems. A 2025 IEEE systematic review found agentic AI being applied across diagnostics, autonomous documentation, and clinical training, with success dependent on governance design, not model size. McKinsey's 2025 Technology Trends Outlook identified billing, coding, and prior authorization as lower-risk entry points before advancing to patient-facing workflows.

2.3 Explainability as Clinical Mandate

Black-box models are no longer clinically or legally acceptable. A 2025 systematic review of 70 studies across oncology, cardiology, and clinical decision support confirmed SHAP and LIME as the dominant clinical explainability techniques. XAI enhances clinician trust and decision-making, yet challenges persist around regulatory standardization and inconsistent validation (PMC, 2025).

2.4 RAG as the Hallucination Mitigation Standard

LLM hallucination remains the primary clinical safety risk. Research in *Frontiers in Digital Health* (2026) warns that plausible but incorrect outputs accumulate undetected and erode health information integrity. RAG architecture grounding outputs in verified, citable documents has emerged as the primary mitigation strategy. For payment integrity, this means anchoring recommendations to specific payer-provider contracts and coding policies rather than model training data.

3. Opportunities and Threats for Cotiviti

3.1 Opportunities

- **Payment Integrity Automation:** Agentic AI cross-referencing claims against documented guidelines in real time can identify billing anomalies without manual audit. McKinsey (2024) reports 40-60% reductions in administrative costs for early adopters.
- **Clinical Decision Support at Scale:** ML predictions paired with SHAP explainability can reduce diagnostic errors by up to 45%, directly supporting Cotiviti's treatment analytics products and utilization management workflows.
- **Payer-Side Intelligence:** With payer applications recording the highest CAGR in the agentic AI market, Cotiviti is positioned to lead in AI-driven contract compliance, automated prior authorization, and claims adjudication.
- **Operational Efficiency:** U.S. physicians spent 28 hours per week on administrative tasks in 2024. Agentic systems targeting documentation, scheduling, and coding review reclaim this burden directly (NextGenInvent, 2026).

3.2 Threats

- **Hallucination Risk:** Subtle, plausible-but-incorrect LLM outputs remain a critical safety risk in clinical and payment contexts. Mitigation requires RAG grounding, strict temperature controls, and structured refusals for out-of-scope queries.
- **Regulatory Ambiguity:** While the FDA institutionalized autonomous administrative AI in December 2025, clinical reasoning agents still lack a unified binding regulatory standard in the U.S., creating deployment liability risk.
- **Algorithmic Bias:** Models trained on historically skewed datasets can perpetuate disparities in payment determinations. Without bias auditing, scaling amplifies existing inequities across patient populations.
- **Platform Consolidation:** Gartner projects 33% enterprise agentic AI adoption by 2028, with consolidation around Microsoft and Epic creating barriers for specialized vendors. Cotiviti must differentiate through clinical domain depth.

4. Strategic Recommendations for Cotiviti

4.1 Deploy RAG-Grounded Reasoning for Payment Integrity

Cotiviti should shift from rule-based claim review to RAG-powered agentic systems that cite specific guideline sections supporting or contradicting each claim. This produces auditable, defensible outputs that reduce legal exposure and accelerate dispute resolution. The accompanying POC demonstrates this pattern at proof-of-concept scale using ADA 2023 guidelines; the same architecture applies directly to oncology billing, surgical coding, and payer-provider contract compliance.

4.2 Establish an Internal AI Governance Standard

Before scaling agentic deployments, Cotiviti must mandate: (a) SHAP-based explainability for all ML outputs, (b) RAG grounding with mandatory citation requirements, (c) continuous hallucination monitoring pipelines, and (d) demographic bias testing before production release. McKinsey explicitly recommends this governance-first approach. For Cotiviti, governance is not merely compliance; it is a competitive differentiator in a market where payers and providers require auditable AI.

4.3 Build a Clinical AI Research Internship Pipeline

The talent required building and governing agentic clinical AI researchers combining ML engineering, healthcare domain knowledge, and analytical writing is scarce and increasingly competitive. Cotiviti should establish a structured internship program targeting recent graduates with demonstrated research output in healthcare AI to build institutional knowledge before the talent market tightens further.

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